
From PDE-level gap to algorithmic OGP: what remains to prove?

Utkarsh Utkarsh
MIT
utkarsh5@mit.edu

Theodoros Xenakis
MIT
txenakis@mit.edu

Abstract

This memo contains an overview of what was proved by us [UX26] as part of our class project in Prof. Gamarnik's 15.070 class at MIT, spring 2026. We describe in more detail how this fits in with existing literature, and formulate some questions that could help us recognize the potential novelty and implications of our work. The main purpose at this stage is to ask whether flatness of the $\mathcal{U}$ -mimizer has known algorithmic implications. In particular we would like to know whether our PDE-level (flatness) result is already known, and whether it implies strict separation between $\mathcal{U}$ and $\mathcal{L}$ variational problems.

1 Background

Consider the general spin glass objective of maximizing the Hamiltonian function H_N , for a configuration $\sigma \in \Sigma_N \subset \mathbb{R}^N$. Define

$$\text{OPT}_N := \frac{1}{N} \max\{H_N(\sigma) : \sigma \in \Sigma_N\} \quad (1)$$

Let the Parisi PDE be written as

$$\partial_t \Phi_\gamma(t, x) + \frac{1}{2} \xi''(t) (\partial_x^2 \Phi_\gamma(t, x) + \gamma(t) (\partial_x \Phi_\gamma(t, x))^2) = 0 \quad (2)$$

$$\Phi_\gamma(1, x) = |x| \quad (3)$$

and define the following functional.

$$P(\gamma) := \Phi_\gamma(0, 0) - \frac{1}{2} \int_0^1 t \xi''(t) \gamma(t) dt \quad (4)$$

Auffinger and Chen [AC17] have proved that the following limit holds almost surely.

$$\lim_{N \rightarrow \infty} \text{OPT}_N = \inf_{\gamma \in \mathcal{U}} P(\gamma) \quad (5)$$

where we define the space

$$\mathcal{U} := \left\{ \gamma : [0, 1] \rightarrow \mathbb{R}_{\geq 0} : \gamma \text{ is non decreasing, } \int_0^1 \gamma(t) dt < \infty \right\} \quad (6)$$

Finding $\sigma^* := \arg \max_{\sigma \in \Sigma_N} H_N(\sigma)$ is notably NP-Hard, however, finding a near optimum can be a relatively easier problem. Specifically, in [EAMS21] they define the space of functions ¹

$$\mathcal{L} := \left\{ \gamma : [0, 1] \rightarrow \mathbb{R}_{\geq 0} : \|\xi'' \gamma\|_{\text{TV}[0,1]} < \infty, \forall t \in [0, 1], \int_0^1 \xi''(t) \gamma(t) dt < \infty \right\} \quad (7)$$

and develop a message passing algorithm, that is said to find a near optimal solution within an arbitrary ε -distance from ground state. Mathematically, given an $\varepsilon > 0$, their algorithm achieves

$$H_N(\sigma^{\text{alg}})/N \geq (1 - \varepsilon) \inf_{\gamma \in \mathcal{L}} P(\gamma) \quad (8)$$

As they importantly point out, since $\mathcal{U} \subsetneq \mathcal{L}$ it holds that

$$\inf_{\gamma \in \mathcal{L}} P(\gamma) \leq \inf_{\gamma \in \mathcal{U}} P(\gamma) \quad (9)$$

In their paper, they additionally prove that when the minimizer $\gamma_{\mathcal{U}}^*$ is strictly increasing, we have equality

$$\inf_{\gamma \in \mathcal{L}} P(\gamma) = \inf_{\gamma \in \mathcal{U}} P(\gamma) \quad (10)$$

2 OGP

The assumption that $\gamma_{\mathcal{U}}^*$ is strictly increasing is phrased by the authors of [EAMS21] as a *no-overlap gap assumption*.

Informally, the *overlap gap property* (OGP) means that if one looks at all the near optimal configurations of the Hamiltonian, they are either very close to each other, or very far apart. In the middle there is a "gap" where no valid solutions exist. If OGP does not hold, then solutions can be found at any distance from one another. In formal terms:

Definition 2.1 (OGP). Let

$$\text{OPT}_N = \frac{1}{N} \max_{\sigma \in \{\pm 1\}^N} H_N(\sigma)$$

The model is set to satisfy the Overlap Gap Property if there exists constants $0 < a < b < 1$ and $\varepsilon > 0$ such that with high probability as $N \rightarrow \infty$, for any two (near optimal) configurations satisfying

$$\frac{1}{N} H_N(\hat{\sigma}) \geq \text{OPT}_N - \varepsilon, \quad \frac{1}{N} H_N(\tilde{\sigma}) \geq \text{OPT}_N - \varepsilon,$$

their overlap satisfies

$$R(\hat{\sigma}, \tilde{\sigma}) := \frac{1}{N} \langle \hat{\sigma}, \tilde{\sigma} \rangle \notin [a, b]$$

2.1 PDE-Level Gap

The overlap gap assumption from [EAMS21] - is equivalent to stating that the Parisi order parameter does not have vanishing mass, which in a sense is a kind of *PDE-level OGP*. Specifically, this is also what we define in [UX26]

$$\begin{aligned} \gamma^* \text{ is strictly increasing} &\iff \text{No PDE level OGP} \\ \gamma^* \text{ has flat interval} &\iff \text{PDE level OGP} \end{aligned}$$

The main result of our work [UX26], is that for pure spin $p > 2$, the zero-temperature Parisi order parameter $\gamma_{\mathcal{U}}^*$ cannot be strictly increasing - hence we have PDE-level OGP.

PDE level OGP and finite- N binary OGP are very much related, but the second one is stricter.

¹For $f : \mathbb{R} \rightarrow \mathbb{R}$ define the TV-distance on a set J as

$$\|f\|_{\text{TV}(J)} := \sup_n \sup_{t_0 < t_1 < \dots < t_n \in J} \sum_{i=1}^n |f(t_i) - f(t_{i-1})|$$

An initial question that we are interested in answering is if the following holds:

$$\gamma^* \text{ has a flat interval } \stackrel{?}{\Rightarrow} \inf_{\gamma \in \mathcal{L}} P(\gamma) < \inf_{\gamma \in \mathcal{U}} P(\gamma) \quad (11)$$

If the above implication holds, then the extended variational value in [EAMS21] is strictly below the true Parisi ground-state value. Since $\inf_{\gamma \in \mathcal{U}} P(\gamma)$ is the asymptotic optimum, while $\inf_{\gamma \in \mathcal{L}} P(\gamma)$ is the value characterized by the message-passing variational principle in [EAMS21], this would imply a genuine algorithmic gap for this class of message-passing algorithms. I.e. these algorithms cannot asymptotically reach the ground-state energy. This would be an algorithmic consequence of the PDE-level flat interval, although it would still be weaker than a full hardness theorem as the ones presented in [HS22], and would not by itself rule out all polynomial-time algorithms.

2.2 Branching OGP and algorithmic bridge

Classical OGP concerns itself with *pairs* of near-optima. *Branching* OGP, however, which is what [HS22], revolves around, is a higher-order tree-structures version. Instead of considering two solutions, one considers M tuples whose pairwise overlaps from a continuous branching tree. The presence or absence of such trees is the object that turns out to be the algorithmic threshold, not classical pairwise OGP.

The relevance to our flatness result is the following. In the models where it has been established, the branching-OGP threshold is *exactly* the optimum of the Parisi functional over $\mathcal{L}$. Concretely, while the algorithm described in [EAMS21] achieves $\inf_{\gamma \in \mathcal{L}} P(\gamma)$ up to an ε , [HS22] show that no Lipschitz algorithm can do better. Thus $\inf_{\gamma \in \mathcal{L}} P(\gamma)$ is the tight threshold for the entire Lipschitz class. This would potentially upgrade the result from the previous subsection into a general result for all Lipschitz algorithms - i.e. not only the message passing algorithm in [EAMS21].

The caveat is that the original paper [HS22] only applies to *even* p -spin models. In more recent work, Huang and Sellke [HS23] also prove a threshold for odd spins, but in this work they only consider *spherical* spin glasses - whereas we have focused on Ising models.

3 Questions

The reason the following questions matter is that while our current theorem gives a PDE-level gap, it says nothing - explicitly at least - of algorithmic hardness. [EAMS21] shows that strict increase of $\gamma_{\mathcal{U}}^*$ is sufficient for equality between the minimizer in the two different spaces $\mathcal{U}$, $\mathcal{L}$. The key issue is whether this failure forces strict inequality, or whether equality could hold for other reasons. Resolving this could clarify the implications of our work.

- Is the statement "pure $p > 2$ implies no strictly increasing zero-temperature Parisi minimizer" already known explicitly? Is it implicit in [EAMS21], [HS22], or [AC17]?
- If not, is it useful/novel as a short PDE-level argument?
- Suppose $\gamma_{\mathcal{U}}^*$ has a nontrivial flat interval. Should this imply

$$\inf_{\gamma \in \mathcal{L}} P(\gamma) < \inf_{\gamma \in \mathcal{U}} P(\gamma)$$

- If the implication is expected to be true, but unknown, what is the right way to prove it?
- Is it known whether $\inf_{\mathcal{L}} P < \inf_{\mathcal{U}} P$ for pure odd $p \geq 3$? (Especially $p = 3$)
- Is the branching-OGP/ tight Lipschitz hardness machinery available for the Ising odd pure p -spin model, or is it only established for even ([HS22]) and spherical ([HS23]) spins?

References

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